Accuracy is not efficiency: MSE and split-conformal interval width can diverge arbitrarily

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Abstract

A common reflex when building conformal prediction intervals is to first fit the most *accurate* point predictor — lowest mean squared error (MSE) — and then conformalize its residuals, in the belief that a better predictor yields shorter intervals. We make precise the sense in which this reflex is wrong. The width of a split-conformal interval built from absolute residuals is the $(1-\alpha)$-quantile of $|Y - f(X)|$, a functional that is *completely insensitive* to how the residual law behaves above that quantile — exactly the region that dominates MSE. We show (Proposition 1) that, as a consequence, MSE and conformal width can be misaligned by an arbitrary factor in either direction: there is a predictor with arbitrarily large MSE whose valid conformal interval is *degenerate* (zero width), and a predictor with arbitrarily small MSE whose interval is strictly positive. We then identify the correct order (Proposition 2): if the absolute residuals of f_1 are stochastically dominated by those of f_2, then f_1 has both smaller MSE *and* shorter intervals at every level simultaneously — so the folklore fails only because MSE is a lossy scalar summary of that order, not because accuracy and efficiency are genuinely opposed. A corollary pins down the benign regime (a common-shape scale family) where "lower MSE $\Rightarrow$ shorter intervals" is exactly true. A regression simulation realizes the reversal: a model with 15× larger MSE produces intervals 3.4× narrower than a competitor, both at 90% coverage.

1. Setup and notation

Let $(X_i, Y_i)_{i=1}^{n+1}$ be exchangeable, $X_i \in \mathcal{X}$, $Y_i \in \mathbb{R}$. Fix a predictor $f : \mathcal{X} \to \mathbb{R}$ (fit on an independent training fold, so it may be treated as fixed). **Split (inductive) conformal prediction** with the absolute-residual score uses the calibration scores

$$R_i = |Y_i - f(X_i)|, \qquad i = 1, \dots, n,$$

and forms, for a test point x, the interval

$$\widehat{C}(x) = [\, f(x) - \hat{q},\ f(x) + \hat{q}\,], \qquad \hat{q} = R_{(k)}, \quad k = \lceil (1-\alpha)(n+1) \rceil,$$

where $R_{(1)} \le \dots \le R_{(n)}$ are the order statistics (and $\hat{q} = +\infty$ if $k > n$). The interval has half-width $\hat{q}$, so its **width** is $W_n = 2\hat{q}$. By exchangeability, $\widehat{C}$ is *marginally valid* with no assumption on the data law beyond exchangeability [1,2,3]:

$$\mathbb{P}(Y_{n+1} \in \widehat{C}(X_{n+1})) \ \ge\ 1 - \alpha. \tag{1}$$

Let $R = |Y - f(X)|$ denote a generic absolute residual, with CDF F_R and survival function $S_R(t) = \mathbb{P}(R > t)$. Define the quantile $q_{1-\alpha}(R) = \inf\{t \ge 0 : F_R(t) \ge 1 - \alpha\}$. As $n \to \infty$, $\hat{q} \to q_{1-\alpha}(R)$ almost surely (at continuity points), so we study the **population width**

$$W(f) \;=\; 2\,q_{1-\alpha}(|Y - f(X)|), \tag{2}$$

and compare it to the quantity practitioners actually minimize, the **mean squared error**

$$M(f) \;=\; \mathbb{E}\big[(Y - f(X))^2\big] \;=\; \mathbb{E}[R^2]. \tag{3}$$

The crux. The width (2) is a quantile of F_R: it depends on F_R only through its restriction to $[0, q_{1-\alpha}(R)]$. Any reallocation of the upper α probability mass — the part of the residual law lying *above* the quantile — to larger and larger values leaves $W(f)$ unchanged while sending $M(f) = \mathbb{E}[R^2]$ to infinity. MSE is dominated by precisely the tail that conformal width ignores. The rest of the note develops the consequences.

2. Contribution

2.1 Arbitrary misalignment

Proposition 1 (MSE and conformal width are arbitrarily misaligned). Fix $\alpha \in (0,1)$. *(a)* There exists an absolute-residual law with M arbitrarily large yet $W = 0$: a predictor whose conformal interval is **degenerate** (a single point) but still satisfies the validity guarantee (1). *(b)* There exists an absolute-residual law with $W > 0$ and M arbitrarily small. Consequently, for every $B > 0$ there are predictors f_1, f_2 of the same response with

$$M(f_1)/M(f_2) \;\geq\; B \qquad \text{while} \qquad W(f_1) \;<\; W(f_2), \tag{4}$$

and, symmetrically, predictors with the MSE ordering and width ordering both reversed.

Proof. Any nonnegative law is realizable as $|Y - f(X)|$ (e.g. take $f \equiv 0$ and $Y = \pm R$ with the desired law of R), so it suffices to exhibit residual laws.

(a) Let $0 < \delta < \alpha$ and put $R_1 = M \cdot \mathbf{1}\{U < \delta\}$ with $U \sim \mathrm{Unif}(0,1)$; that is, $R_1 = M$ with probability δ and $R_1 = 0$ otherwise. Then $\mathbb{P}(R_1 = 0) = 1 - \delta > 1 - \alpha$, so $F_{R_1}(0) = 1 - \delta \geq 1 - \alpha$ and $q_{1-\alpha}(R_1) = 0$, giving $W(f_1) = 0$. But $M(f_1) = \mathbb{E}[R_1^2] = \delta M^2 \to \infty$ as $M \to \infty$. The interval $\widehat{C}(x) = \{f_1(x)\}$ is degenerate yet, by (1), covers with probability $\geq 1 - \alpha$: the catastrophic errors occur with probability $\delta < \alpha$, within the miscoverage budget.

(b) Let $R_2 \equiv c$ be constant. Then $q_{1-\alpha}(R_2) = c$, so $W(f_2) = 2c > 0$, while $M(f_2) = c^2 \to 0$ as $c \to 0$.

Combining, $M(f_1)/M(f_2) = \delta M^2/c^2$, which exceeds any B for M large and c small, while $W(f_1) = 0 < 2c = W(f_2)$, giving (4). Exchanging the roles of "bulk" and "tail" between the two laws reverses both orderings. ■

The degenerate case in part *(a)* is the sharpest possible statement: *no* finite MSE penalty can force a conformal interval to be wide, because conformal simply declines to cover the small-probability region where the model fails. (This is also a sharpening of the well-known gap between marginal and conditional coverage; see §4.)

2.2 The right order

What ordering of predictors *does* guarantee shorter intervals? Not the second moment, but the whole upper tail — i.e. the stochastic order on absolute residuals.

Proposition 2 (stochastic dominance is the right order). Let $R_1 = |Y - f_1(X)|$ and $R_2 = |Y - f_2(X)|$, and suppose $R_1 \preceq_{\text{st}} R_2$, i.e. $S_{R_1}(t) \le S_{R_2}(t)$ for all $t \ge 0$. Then *(i)* $q_{1-\alpha}(R_1) \le q_{1-\alpha}(R_2)$ for **every** $\alpha \in (0,1)$, so $W(f_1) \le W(f_2)$ at all coverage levels simultaneously; and *(ii)* $M(f_1) \le M(f_2)$.

Proof. *(i)* $S_{R_1} \le S_{R_2}$ means $F_{R_1} \ge F_{R_2}$ pointwise, so $\{t : F_{R_2}(t) \ge 1-\alpha\} \subseteq \{t : F_{R_1}(t) \ge 1-\alpha\}$; taking infima gives $q_{1-\alpha}(R_1) \le q_{1-\alpha}(R_2)$. *(ii)* For a nonnegative variable, $\mathbb{E}[R^2] = \int_0^\infty \mathbb{P}(R^2 > u)\,du = \int_0^\infty 2t\,S_R(t)\,dt$ (substitute $u = t^2$). This is monotone in the survival function, so $S_{R_1} \le S_{R_2}$ gives $M(f_1) \le M(f_2)$. ■

Proposition 2 reframes the folklore. "Model 1 is more accurate" should mean $R_1 \preceq_{\text{st}} R_2$ — its residual is *stochastically smaller*. Under that (strong, level-free) notion, accuracy and conformal efficiency never conflict; lower MSE and shorter intervals come together. The reversals of Proposition 1 are possible *only* when neither residual law dominates the other, so that the scalar $\mathbb{E}[R^2]$ and the scalar $q_{1-\alpha}(R)$ — two different one-dimensional projections of the same law — disagree about which is "smaller."

2.3 When the folklore is exactly right

The misalignment requires the two residual *shapes* to differ. Within a single shape it cannot happen.

Corollary (common-shape scale family). Suppose $R_i \overset{d}{=} s_i Z$ for $i = 1, 2$, where $Z \ge 0$ is a fixed shape with $\mathbb{E}[Z^2] < \infty$ and $s_i > 0$ are scales. Then

$$M(f_i) = s_i^2\,\mathbb{E}[Z^2], \qquad W(f_i) = 2s_i\,q_{1-\alpha}(Z),$$

so $M(f_1) \le M(f_2) \iff s_1 \le s_2 \iff W(f_1) \le W(f_2)$ at every α.

Proof. Both M and W are increasing in s_i through the same parameter; $q_{1-\alpha}(sZ) = s\,q_{1-\alpha}(Z)$ by positive homogeneity of quantiles. ■

So if two predictors have residuals of the same shape differing only in scale — e.g. homoskedastic Gaussian errors of different variances — minimizing MSE is exactly minimizing conformal width. The cautionary tale of Proposition 1 lives entirely in the gap between shapes.

3. Simulation

`sim.py` (seed `20260601`) realizes a concrete, non-degenerate version of Proposition 1. The DGP is $X \sim \text{Unif}(0,1)$, $Y = X + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.05^2)$; the Bayes predictor is $f^*(x) = x$. We compare two fixed predictors:

- **Model A ("occasionally catastrophic")**: $f_A(x) = x$ except on a region of probability $\delta = 0.04 < \alpha = 0.10$ (here $x \in [0.48, 0.52]$), where $f_A(x) = x + 5$.
- **Model B ("uniformly biased")**: $f_B(x) = x + 0.25$ everywhere.

We run split conformal at $\alpha = 0.10$ with $n_{\text{cal}} = 500$, evaluate on 4000 test points, and average over 300 fresh draws ($\pm$ below is SD across draws). MSE and the population quantiles are computed on a separate 4×10^5-point sample.

quantity	Model A (spiky)	Model B (biased)
MSE	**1.0043**	**0.0650**
population 0.9-quantile of $\lvert r \rvert$	0.0932	0.3140
empirical coverage	0.9003 ± 0.0132	0.8995 ± 0.0144
interval width	**0.1868** ± 0.0095	**0.6281** ± 0.0075

Model A has 15.5× the MSE of Model B, yet its intervals are 3.4× *narrower* (0.187 vs 0.628), and both hold 90% marginal coverage. Figure 1 shows why: the conformal width is read off where each residual CDF crosses 0.9. Model A's CDF crosses early (small quantile $\hat{q}_A \approx 0.09$) and then carries a flat fat tail out to $r \approx 5$ — that tail is what makes its MSE large, but it sits *above* the 0.9-quantile and so never touches the width. Model B has no tail but a higher bulk, hence a larger quantile $\hat{q}_B \approx 0.31$.

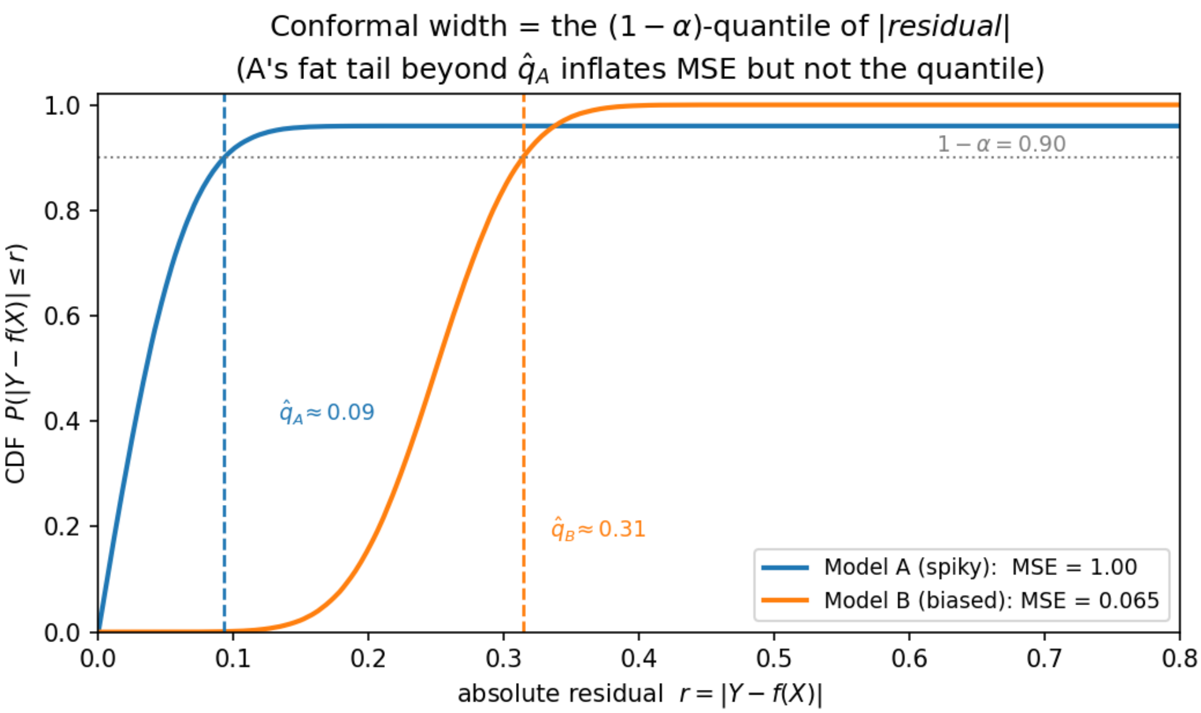

Figure 1: Conformal width is the (1-alpha)-quantile of the absolute residual; Model A's fat tail beyond its quantile inflates MSE but not interval width.

Figure 2 shows the two bands over the data. Model A's band is far tighter and is valid *marginally* — but on its bad region its conditional coverage is exactly 0.000 ± 0.000: the interval is $f_A(x) \pm 0.09$ around a prediction that is wrong by 5. Marginal validity buys the narrow interval by spending the entire α-budget in one place.

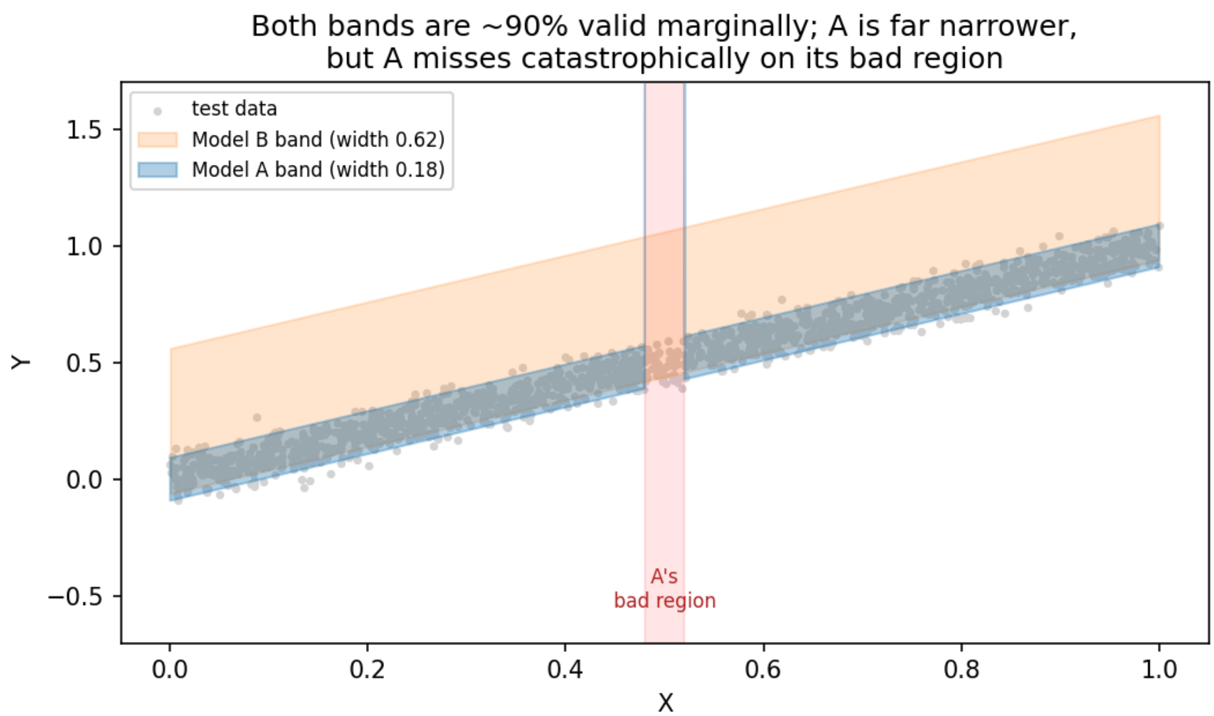

Figure 2: Both bands are ~90% valid marginally; Model A is far narrower but misses catastrophically on its bad region (where its band leaves the frame).

4. Discussion

Practical reading. If you will conformalize, do not select the base predictor — or the training objective — by MSE/RMSE. Equation (2) says the only thing that matters for width is the $(1-\alpha)$-quantile of $|Y - f(X)|$, so the right target is that quantile (or, asymmetrically, the conditional quantiles of Y). This is exactly the rationale for **conformalized quantile regression** [4], which trains with the pinball loss instead of squared loss; Proposition 1 gives the population-level reason such methods can dominate residual-of-the-mean conformal, and the Corollary explains why the difference vanishes for homoskedastic same-shape errors. A clean operational consequence: when choosing among candidate predictors to conformalize, rank them by a held-out estimate of $q_{1-\alpha}(|Y - f(X)|)$, not by validation MSE — the two can disagree about the winner.

Relation to marginal vs. conditional coverage. The degenerate construction in Proposition 1(a) is also an extreme instance of the known gap between marginal and conditional validity [3,6]: a marginally valid interval can have arbitrarily bad coverage on a subpopulation. Here the subpopulation is Model A's bad region, where conditional coverage is 0. The two phenomena are the same coin: width-insensitivity to the upper-α tail *is* the freedom to abandon an α-fraction of inputs.

Relation to prior work. That MSE-trained mean predictors yield non-adaptive, often inefficient conformal intervals is folklore in the conformal literature and the stated motivation for adaptive scores [4,7] and gentle-introduction treatments [5]. Our contribution is to make the failure *quantitative and two-directional* (Proposition 1: misalignment by an arbitrary factor, including a zero-width interval at unbounded MSE), and to identify the *exact* order that removes it (Proposition 2: stochastic dominance of $|residual|$) together with the precise benign regime (the Corollary). We did not find this order-theoretic framing — width as a tail-insensitive quantile functional, stochastic dominance as the level-uniform notion of accuracy, scale families as the equivalence regime — stated explicitly elsewhere; standard references on stochastic orders [8] supply the underlying facts.

Limitations.

- *Population idealization.* (2) is the $n \to \infty$ width; finite n adds $O(n^{-1/2})$ fluctuation and a mild upward bias from (1). This does not affect the ordering message, and the simulation uses finite n and confirms it.
- *Symmetric absolute-residual score.* We analyze the standard symmetric score. Locally-adaptive (normalized) scores and CQR change *which* quantile is relevant but not the structural fact that width is a quantile of a score and hence tail-insensitive; the same misalignment logic applies to the normalized score. Asymmetric or set-valued scores would need a separate (analogous) treatment.
- *Fixed predictors.* We compare given f_1, f_2; in practice the training procedure shapes the residual law itself. The point is structural — what conformal *sees* is a quantile — not a claim that one cannot train for it.
- *Boundary, not blanket, result.* Proposition 1 uses heavy/degenerate residual shapes. The Corollary shows that in benign same-shape settings MSE is a perfectly good proxy. The message is "MSE can mislead arbitrarily across shapes," not "MSE is always misleading."

References

1. V. Vovk, A. Gammerman, G. Shafer. *Algorithmic Learning in a Random World.* Springer, 2005. doi:10.1007/b106715.
2. H. Papadopoulos, K. Proedrou, V. Vovk, A. Gammerman. "Inductive Confidence Machines

for Regression." *ECML 2002*, LNCS 2430, 345–356. doi:10.1007/3-540-36755-1_29.

3. J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman. "Distribution-Free Predictive Inference for Regression." *JASA* 113(523):1094–1111, 2018. arXiv:1604.04173.
4. Y. Romano, E. Patterson, E. J. Candès. "Conformalized Quantile Regression." *NeurIPS 2019.* arXiv:1905.03222.
5. A. N. Angelopoulos, S. Bates. "Conformal Prediction: A Gentle Introduction." *Foundations and Trends in Machine Learning* 16(4):494–591, 2023. arXiv:2107.07511.
6. R. Foygel Barber, E. J. Candès, A. Ramdas, R. J. Tibshirani. "The limits of distribution-free conditional predictive inference." *Information and Inference* 10(2):455–482, 2021. arXiv:1903.04684.
7. M. Sesia, E. J. Candès. "A comparison of some conformal quantile regression methods." *Stat* 9(1):e261, 2020. arXiv:1909.05433.
8. M. Shaked, J. G. Shanthikumar. *Stochastic Orders.* Springer, 2007. doi:10.1007/978-0-387-34675-5.